

Supplementary Information

CrossPiezo: Elite-Tail Instability in Cross-Database Piezoelectric Screening

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1 Supplementary notes

1.1 Portfolio estimand definition (Note 1)

The portfolio benchmark reports two related quantities. *Worst-source recall* is a material-level (micro) metric: it counts how many of the top- q^*n materials selected by a strategy are also present in the JARVIS and MP elite sets, divided by q^*n , with no group weighting. *Paired difference in recall* is the difference between the material-level worst-source recall of the strategy and that of the better single-source baseline, computed on the full observed panel.

The confidence interval reported in Table S3 uses a *full-procedure grouped bootstrap*. Each replicate: (i) resamples reduced-formula groups with replacement and assigns distinct bootstrap identities to duplicated occurrences; (ii) recomputes source-specific rankings and top- q^* elite sets on the bootstrap sample; (iii) re-runs the portfolio strategy and both single-source baselines; (iv) computes the improvement over the better single source within that replicate. The reported 95% interval is the percentile bootstrap interval of these replicate differences. The point estimate is fixed on the full-panel selections and is exactly the arithmetic difference of the Recall column; the bootstrap captures uncertainty from group resampling and re-selection.

The “better single source” is determined per replicate for the interval; the main-text point estimate uses the full-panel better source (here JARVIS-only and MP-only are tied on P0 F1 at $q^* = 10\%$, $b = 1.0$). The portfolio analysis is presented as a risk-management illustration, not as a validated method comparison or as evidence of physical truth.

1.2 Structure matching and manual audit (Note 2)

The CrossPiezo-Invariant-v1 panel was constructed from relaxed CIFs shared by the Materials Project (MP) and JARVIS. The matching pipeline is: (i) load all processed MP and JARVIS piezoelectric records; (ii) compute structure-level overlaps by compositional reduced formula and relaxed-lattice similarity; (iii) apply a frozen similarity threshold to obtain candidate pairs; (iv) perform a response-blind manual audit of the top-matched candidates using a frozen rule set. The audit was response-blind: the auditor did not see piezoelectric response values when deciding whether a candidate pair represented the same or a distinct structure. All audit decisions, exclusion reasons and ambiguous cases are recorded in the frozen panel-membership file.

The frozen matcher tolerances are: fractional length tolerance $\ell_{\text{tol}} = 0.2$, site tolerance $s_{\text{tol}} = 0.3 \text{ \AA}$, angle tolerance 5° , primitive-cell reduction enabled, and lattice scaling enabled (see “configs/matching.yaml”). P0 ($n = 573$) is the full set of accepted structure matches; P2 ($n = 207$) is the subset passing tighter RMS-distance and lattice-distance thresholds ($\text{RMS}_{\text{thr}} \approx 0.0078 \text{ \AA}$, lattice-distance threshold ≈ 0.0036). Table 6 gives the resulting distance distributions. Figure 5 shows that relaxed-structure distance is only weakly associated with cross-source F1 rank discrepancy (Spearman $\rho \approx -0.09$), so large response disagreement is not simply an artefact of loose structure matching.

1.3 Tensor invariants and conventions (Note 3)

All piezoelectric response values are piezoelectric stress tensors $\mathbf{e}$ in C/m^2 with full Cartesian $3 \times 3 \times 3$ components. The three coordinate-invariant scalars are:

- F1, Cartesian Frobenius norm $\|\mathbf{e}\|_F$;
- F3, true collinear longitudinal maximum $\max_{\|\mathbf{n}\|=1} |n_i e_{ijk} n_j n_k|$ computed by numerical optimisation over the unit sphere;

- F4, Kelvin/Mandel operator norm on symmetric strain, computed from the Mandel representation of the 6×3 piezoelectric tensor.

Voigt-to-Cartesian conversion follows the standard convention $11 \rightarrow xx$, $22 \rightarrow yy$, $33 \rightarrow zz$, $23 \rightarrow yz$, $13 \rightarrow xz$, $12 \rightarrow xy$ with factor-of-two handling for shear components as appropriate for the stress tensor. No sign conventions, unit conversions or coordinate rotations were applied silently; all transformations are recorded in the repository transformation history.

1.4 Bootstrap, quantile and tie handling (Note 4)

Screening-resolution curves are evaluated at quantiles $q = 1\%, 2\%, \dots, 50\%$. The chance-adjusted Jaccard overlap is $\tilde{J}_q = (J_q - \mathbb{E}[J_q]) / (1 - \mathbb{E}[J_q])$, where $\mathbb{E}[J_q]$ is the exact hypergeometric expectation under independent rankings. Simultaneous 95% confidence bands for the screening-resolution curve use a paired bootstrap with a studentized sup-norm critical value; this bootstrap resamples matched material pairs (rows) because the curve itself is a cross-sectional function of the ranked universe and the band is constructed for fixed quantiles.

For property-control comparisons and the portfolio benchmark, bootstrap resampling is by reduced-formula group to respect the dependence among entries sharing a composition. In the portfolio full-procedure bootstrap, duplicated groups create distinct bootstrap identities so that re-selection and elite-set sizes are computed on the expanded bootstrap sample. The random seed, number of replicates, studentization details, tie handling and k rounding are recorded in the Phase 7C configuration and result files. Ties in source rankings are broken by stable sorting; duplicate reduced formulae are retained in the ranking but noted in the panel metadata. The persistent-onset criterion requires the lower confidence bound to exceed a threshold δ for at least five consecutive quantiles.

1.5 Source workflow and tensor conventions (Note 5)

Both JARVIS and MP report the piezoelectric stress tensor $\mathbf{e}$ in C/m² using the same Voigt order (xx, yy, zz, yz, xz, xy) and the same engineering-shear factor-of-two convention. JARVIS values are expressed in the source-structure Cartesian frame; MP values are expressed in the MP IEEE-oriented structure frame. Before any response comparison, MP structures are aligned to JARVIS structures (or vice versa) with a frozen “pymatgen” “StructureMatcher” configuration and the recovered rotation is applied to the MP Cartesian tensor. The three reported scalars (F1, F3, F4) are coordinate-frame invariants, so they are unaffected by the source-specific structure-frame choice.

Several calculation settings are not documented in the release metadata used by CrossPiezo: exchange-correlation functional, pseudopotential family, DFPT vs finite-difference ionic response, plane-wave cutoff, k-point density, relaxation thresholds, primitive vs conventional defaults, and sign-convention history. These items are therefore reported as unknown rather than assumed identical (see `docs/mp_vs_jarvis_conventions.md`). The repository convention audit (`scripts/run_convention_audit.py`) confirms that F1, F3 and F4 are rotation invariant, that Voigt/Cartesian conversion and engineering-shear handling are internally consistent, and that the F3 optimiser converges to a dense-grid brute-force oracle. This audit rules out low-level convention errors in the CrossPiezo pipeline; it does not prove that JARVIS and MP use identical calculation settings.

1.6 Screening resolution vs conventional ranking diagnostics (Note 6)

Screening resolution is reported through the chance-adjusted Jaccard curve $\tilde{J}_q$ and its area (nAUCC) because these metrics focus on the elite tail where cross-database disagreement is largest. For comparison, Table 5 and Figures 3–4 report conventional diagnostics: plain Jaccard J_q , chance-adjusted Jaccard $\tilde{J}_q$, overlap coefficient, precision/recall at k , Rank-Biased Overlap (RBO) with $p = 0.95$, top-weighted Kendall τ on the union of the two top- k sets, and top-weighted Spearman ρ on the same union.

These diagnostics illustrate that two rankings can have high global correlation yet disagree on the top few percent of candidates. The synthetic example in Figure 3 constructs two length-100 rankings with global Kendall $\tau \approx 0.98$ but disjoint top-5% sets; global metrics are near-perfect while top-5% precision and Jaccard are zero. On P0 F1 the same pattern holds: global Kendall τ is high (see Table 5), but elite-tail agreement is near chance for small q .

2 Supplementary figures

3 Supplementary tables

Table 1: Corrected high-response sensitivity for P0 F1 at $f = 0.10$ and $f = 0.25$.

Method & f & N & au & CI & nAUCC & CI						
Dual-high	10%	58	0.035	[-0.193, 0.239]	-0.046	[-0.118, 0.044]
JARVIS anchor	10%	57	0.209	[-0.062, 0.374]	0.053	[-0.061, 0.191]
MP anchor	10%	57	-0.053	[-0.235, 0.137]	-0.018	[-0.104, 0.090]
Pooled (exploratory)	10%	58	–	[–, –]	-0.155	[–, –]
Dual-high	25%	143	0.070	[-0.046, 0.181]	0.031	[-0.034, 0.102]
JARVIS anchor	25%	143	-0.019	[-0.141, 0.105]	-0.006	[-0.070, 0.070]
MP anchor	25%	143	0.039	[-0.083, 0.154]	0.007	[-0.052, 0.078]
Pooled (exploratory)	25%	144	–	[–, –]	-0.100	[–, –]

Table 2: Property-control comparison on P0. The same-field copy flag is raised when more than 5% of finite matched pairs have identical JARVIS/MP values.

Attribute & N & au & 95% CI & nAUCC & 95% CI & $\Delta\tau$ & Δ nAUCC & Copy flag
Volume & 573 & 0.967 & [0.963, 0.971] & 0.918 & [0.901, 0.936] & 0.722 & 0.813 & ✓
Band gap & 573 & 0.908 & [0.890, 0.923] & 0.897 & [0.864, 0.908] & 0.663 & 0.792 & ✓
Energy above hull & 573 & 0.611 & [0.547, 0.670] & 0.533 & [0.455, 0.592] & 0.366 & 0.428 & ✓
Dielectric trace & 573 & 0.482 & [0.418, 0.547] & 0.286 & [0.228, 0.344] & 0.236 & 0.181 & ✓

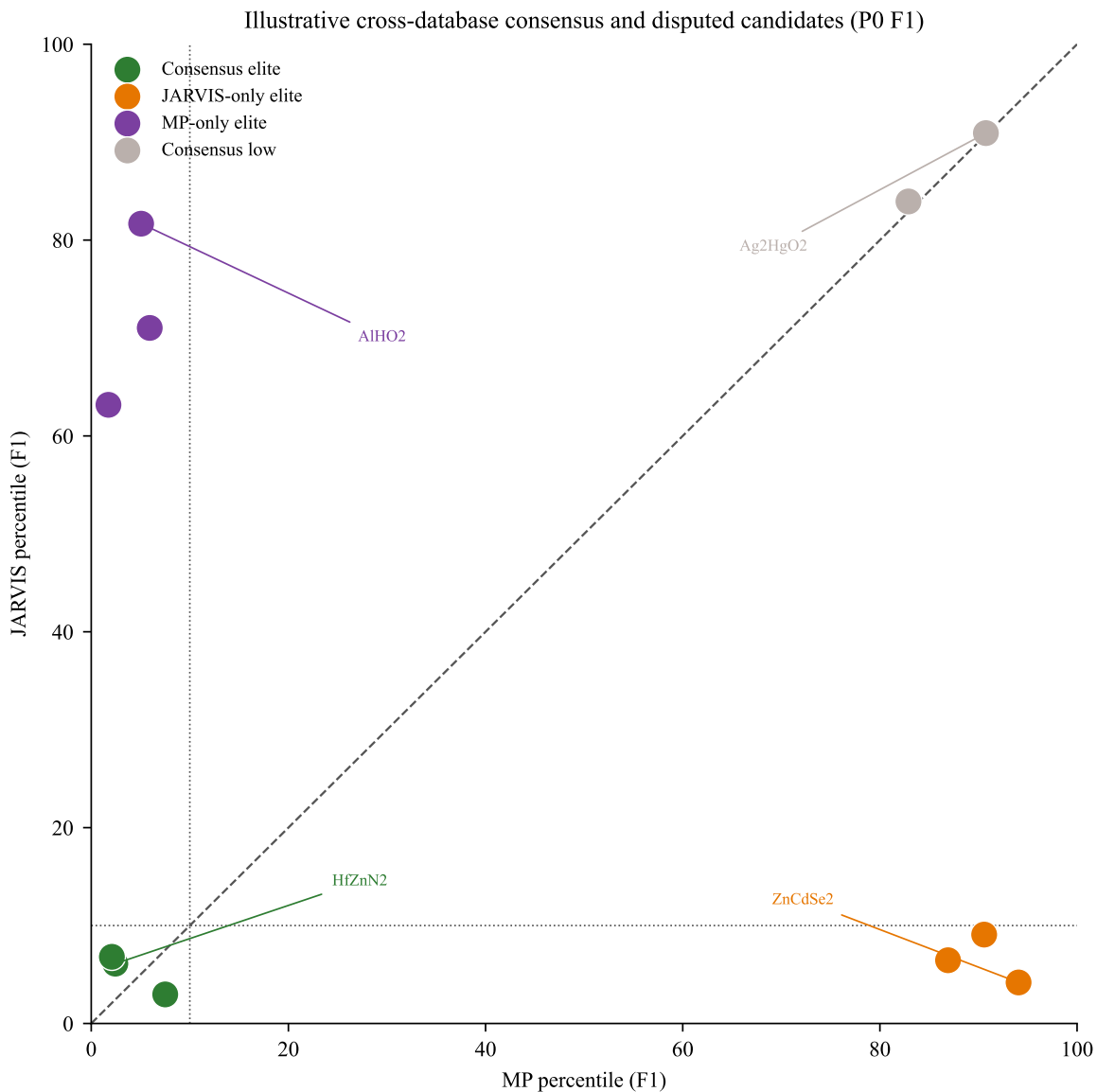


Figure 1: Illustrative cross-database consensus and disputed candidates on P0 F1. Points show JARVIS versus MP percentiles for consensus elite (green), JARVIS-only elite (orange), MP-only elite (purple) and consensus low (grey) materials. The dashed diagonal marks equal ranking; dotted lines mark the 10% elite threshold. These examples are illustrative and are not adjudicated as physically correct or incorrect.

Portfolio cost-coverage frontier on P0 F1 ($q^* = 10\%$)

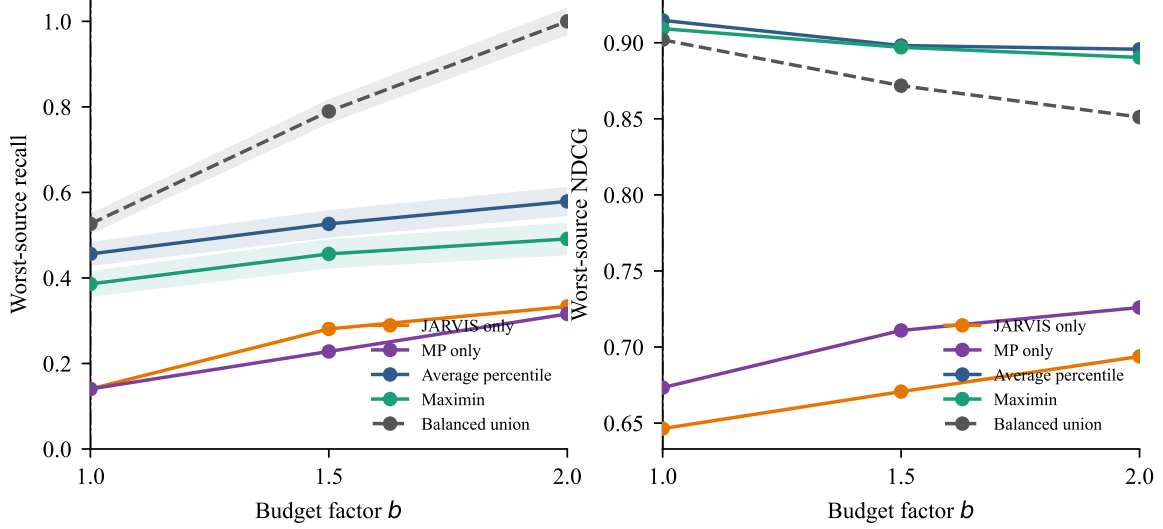


Figure 2: Portfolio cost-coverage frontier on P0 F1 ($q^* = 10\%$). Worst-source recall (left) and NDCG (right) are plotted against budget factor b for the best single-source baselines and three source-robust aggregation strategies. The vertical dashed line marks the equal-budget analysis at $b = 1.0$. Aggregation changes the coverage-quality trade-off but does not validate physical truth. See Note 1 for the distinction between material-level recall and group-mean paired differences.

Table 3: Equal-budget ($b = 1.0$) portfolio results on P0 F1 ($q^* = 10\%$). Paired differences are versus the better single-source baseline and are computed as material-level worst-source recall differences using a full-procedure grouped bootstrap: each replicate resamples reduced-formula groups with replacement, assigns distinct identities to duplicated occurrences, re-runs the portfolio strategy and both single-source baselines, and recomputes the improvement (see Note 1).

Strategy	& Recall	& NDCG	& Coverage	& Δ Recall	& 95% CI
JARVIS only	& 0.140	& 0.646	& 0.099	& 0.000	& [0.000, 0.000]
MP only	& 0.140	& 0.673	& 0.099	& 0.000	& [0.000, 0.000]
Average percentile	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Borda	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Maximin	& 0.386	& 0.909	& 0.099	& 0.246	& [0.107, 0.286]
Intersection first	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Balanced union	& 0.526	& 0.902	& 0.099	& 0.386	& [0.273, 0.441]
Disagreement abstention	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Minimax oracle	& 0.474	& 0.915	& 0.099	& 0.333	& [0.250, 0.414]

Global Kendall $\tau = 0.982$, precision@5% = 0.00, tail $\tau = -0$

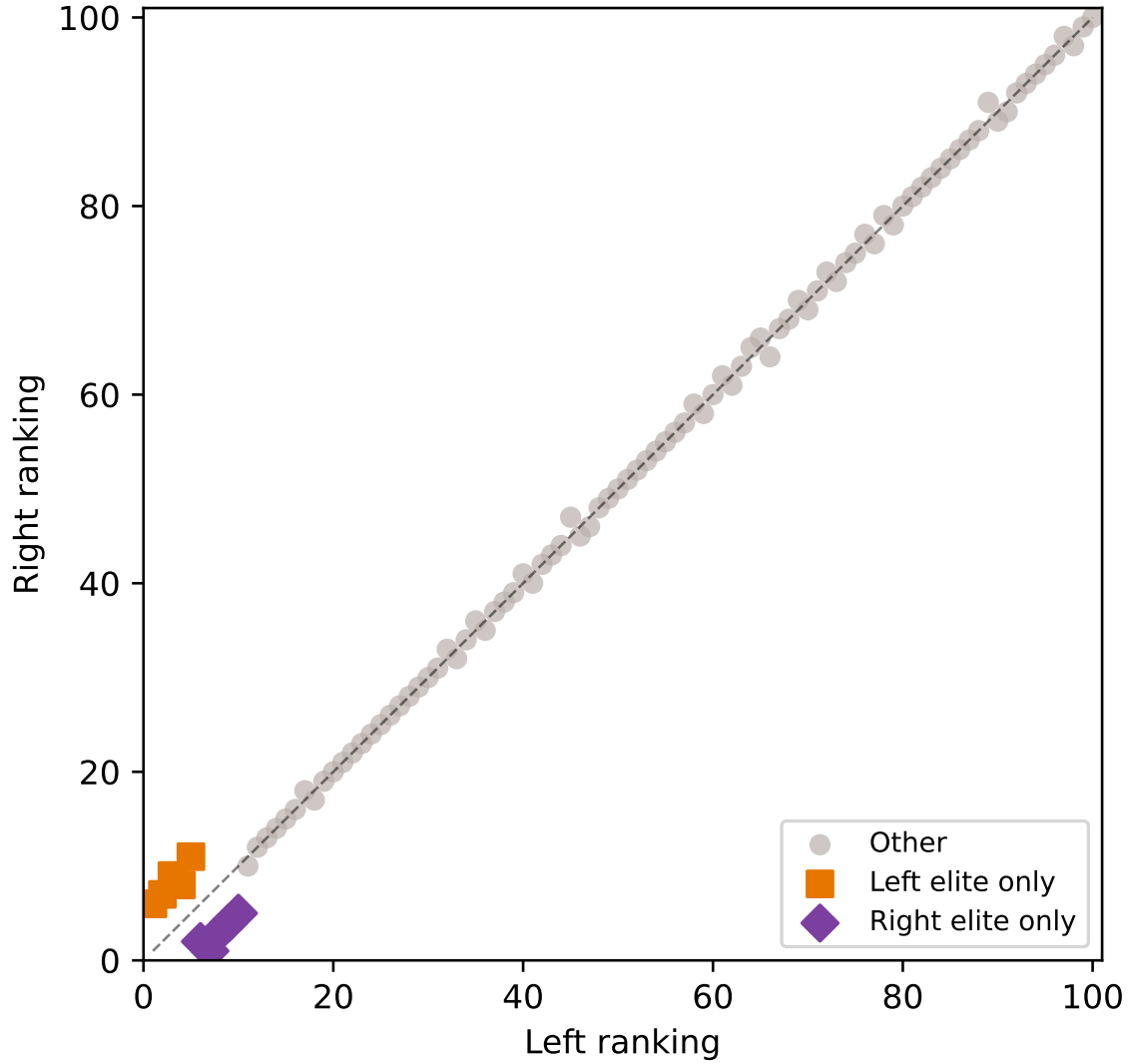


Figure 3: Synthetic motivating example for screening resolution. Two length-100 rankings are constructed with global Kendall $\tau \approx 0.98$ but disjoint top-5% sets, showing that high global rank correlation does not guarantee agreement on elite-screening decisions. The reported conventional metrics (global τ , Spearman ρ , Rank-Biased Overlap) are near-perfect while top-5% precision and Jaccard are zero. See Note 6 for the full metric definitions.

Table 4: Portfolio worst-source recall on P0 F1 across budget factors $b = 1.0, 1.5, 2.0$.

Strategy & $b = 1.0$ & $b = 1.5$ & $b = 2.0$
JARVIS only & 0.140 & 0.281 & 0.333
MP only & 0.140 & 0.228 & 0.316
Average percentile & 0.456 & 0.526 & 0.579
Maximin & 0.386 & 0.456 & 0.491
Balanced union & 0.526 & 0.789 & 1.000

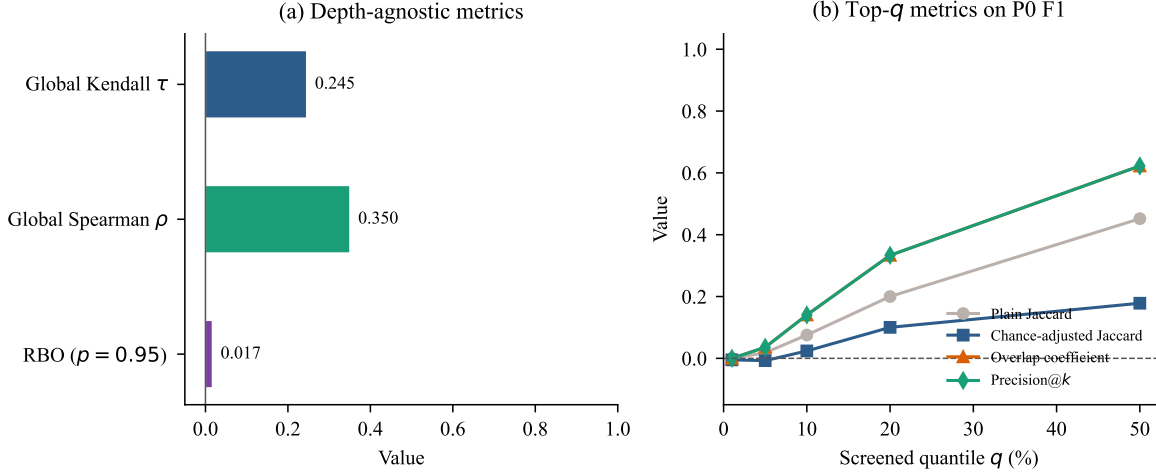


Figure 4: Comparison of conventional ranking diagnostics and screening-resolution on P0 F1. Top panels report global and top- q Kendall τ and Spearman ρ ; bottom panels report chance-adjusted Jaccard $\tilde{J}_q$, plain Jaccard J_q , precision@ k , and Rank-Biased Overlap. Global correlation is high, but elite-tail agreement is near-chance for small q . See Table 5 for the numerical values and Note 6 for definitions.

Table 5: Baseline ranking diagnostics on P0 F1 (global Kendall $\tau = 0.245$, global Spearman $\rho = 0.350$). Columns are, from left to right: screened quantile q , corresponding top- k set size, plain Jaccard, chance-adjusted Jaccard $\tilde{J}_q$, overlap coefficient, precision/recall at k (equal for paired rankings), Rank-Biased Overlap with $p = 0.95$, top-weighted Kendall τ on the union of the two top- k sets, top-weighted Spearman ρ on the same union, and the partial nAUCC of the band containing q (elite 1–10, intermediate 10–20, broad 20–50).

q (%)	k	Plain J	$\tilde{J}_q$	Overlap	Prec@ k	RBO _{0.95}	τ_q	ρ_q	Band nAUCC
1	5	0.000	-0.005	0.000	0.000	0.017	-0.289	-0.588	0.003
5	28	0.018	-0.007	0.036	0.036	0.017	-0.532	-0.760	0.003
10	57	0.075	0.024	0.140	0.140	0.017	-0.386	-0.613	0.003
20	114	0.200	0.100	0.333	0.333	0.017	-0.331	-0.489	0.076
50	286	0.452	0.178	0.622	0.622	0.017	-0.080	-0.112	0.145

Table 6: Frozen structure-matching audit summary. P0 is the full relaxed-structure matched universe; P2 is the tighter subset retained by the RMS and lattice thresholds recorded in “panel_counts.json”.

Panel	Count	Expected	RMS mean	RMS med.	RMS max	Max mean	Max med.	Max max	Lat. mean	Lat. med.	Lat. max
P0	573	573	0.017	0.009	0.168	0.025	0.013	0.298	0.066	0.004	1.897
P2	207	207	0.003	0.003	0.008	0.004	0.004	0.019	0.001	0.001	0.004

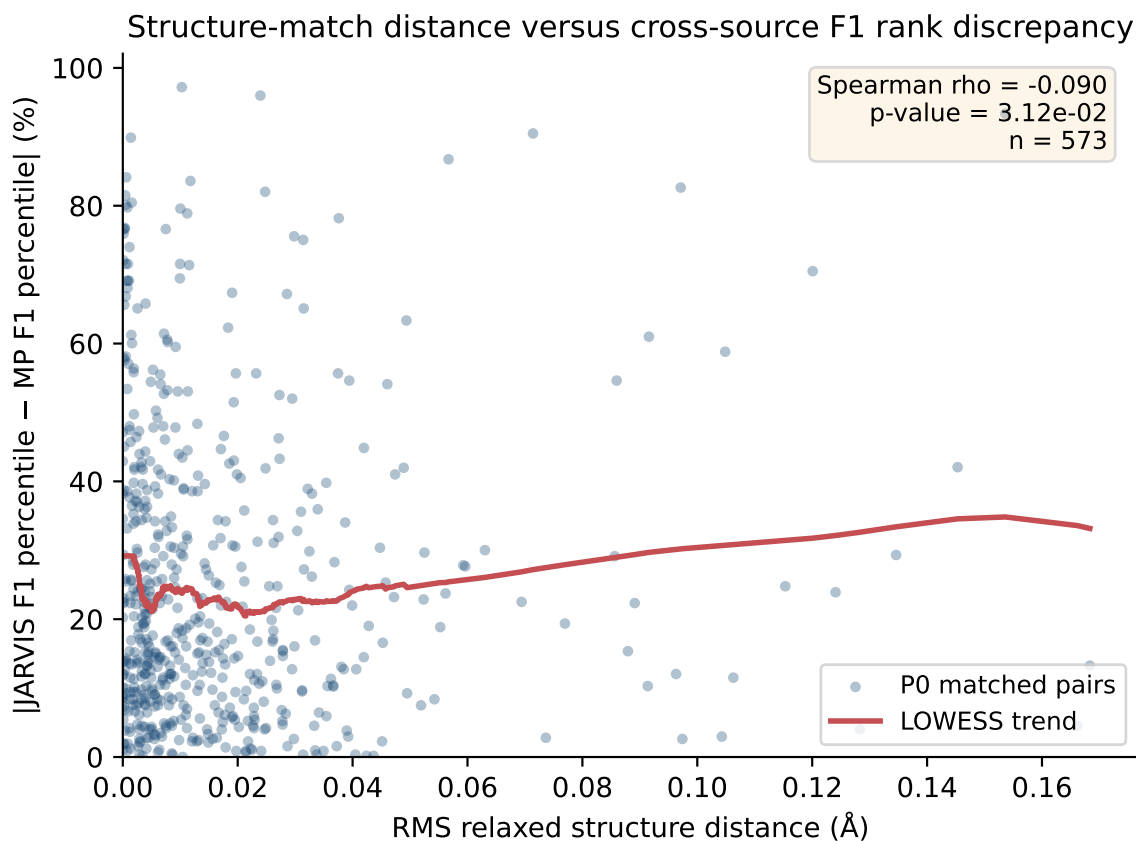


Figure 5: Relaxed-structure RMS distance versus absolute F1 percentile gap for the 573 P0 matched pairs. The weak Spearman correlation ($\rho \approx -0.09$) indicates that large cross-database response disagreement is not dominated by poorly matched structures. The LOWESS curve is illustrative and is not a fitted model.